FIELDCOMMS

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1. Overview & What's Included

FieldComms is a complete off-grid emergency communications server for Raspberry Pi. It provides a Wi-Fi access point (EMCOMM-NET) that any phone, tablet, or laptop on scene can connect to, then access all EmComm tools through a web browser — no internet required. When Ethernet internet is available, live features such as NWS weather alerts, HF propagation data, and APRS-IS automatically activate.

COMPONENT	DESCRIPTION	COMPONENT	DESCRIPTION
30 Web Pages	Full dashboard, net logs, ICS forms, maps, roster, cheat sheets	Dead Man's Switch	Per-net inactivity monitor with warning and trigger states
Net Control Logger	Amateur + Starcom nets, FCC autofill, ICS-309 export	Pre-Flight Checklist	GO/CAUTION/NO-GO deployment readiness check
ICS Platform	Command, Operations, Planning, Logistics, Finance sections	HF Propagation	Solar indices, band conditions, A/K-index
APRS Tactical Map	Graywolf + YAAC merged, offline tiles, overlays	Repeater Database	RepeaterBook CSV, filter by band, mode, ARES affiliation
Member Roster	MCESV/MCEMA directory with certs, equipment, activations	Reference Library	Upload and serve field reference docs across EMCOMM-NET
NTS Radiogram	ARRL-formatted radiogram generator with traffic log	Kiwix Library	WikiMed, Wikipedia, iFixit — offline at port 8081
ICS-213/214/309	Fillable ICS forms with print output and incident archiving	FCC Callsign Lookup	~800K licensees — instant offline search
Winlink Form Import	Import Winlink Express XML forms to incident archive	Pat Winlink	Browser-based backup Winlink client at port 8090
JS8Call Integration	Dashboard card opens JS8Call web API on Windows laptop	ICS Planning P	Interactive 15-phase planning cycle guide

2. Hardware Requirements

Supported Hardware — Pi Server

CONFIGURATION	STORAGE	NOTES
Pi 5 (16 GB) + Pironman MAX 5 enclosure	2× 1 TB NVMe SSD RAID 1 mirror	Tower form factor with OLED display, active cooling, dual M.2 NVMe slots. Two 1 TB SSDs configured as RAID 1 — drives mirror each other for data redundancy. If one SSD fails, the system keeps running on the other with no data loss.

Network Hardware

Wi-Fi is provided by the ASUS RT-BE58 Go travel router — the Pi does not run a Wi-Fi hotspot. The UniFi switch provides wired 2.5 GbE ports for the Pi and other fixed devices.

DEVICE	ROLE	NOTES
ASUS RT-BE58 Go	Wi-Fi 7 access point + DHCP server	Dual-band Wi-Fi 7 (802.11be), 2.5G LAN port, USB-C 18W power.
UniFi Switch Flex 2.5G-5	5-port 2.5 GbE managed switch	PoE powered. Ports: 1=uplink to ASUS, 2=Pi, 3=laptop, 4-5=spare.

Accessories & Cables

ITEM	REQUIRED?	PURPOSE
USB-A drive (32 GB+, labeled FIELDCOMMS)	Recommended	Auto-backup trigger — insert to back up all runtime data instantly
External USB hard drive 1 TB+ (e.g. LaCie Rugged or LaCie Mobile Drive)	Recommended	Incident archive and full system backup. LaCie Rugged is shock/drop/crush resistant. Label FIELDCOMMS for auto-backup compatibility.
USB GPS receiver (u-blox or GlobalSat)	Optional	Live position for tactical map and NWS alerts
USB TNC — Digirig Mobile or SignaLink USB	Optional	Required for APRS transmit/receive via Graywolf or YAAC. FieldComms assigns a stable /dev/tnc0 device name via udev.
Powered USB hub (4- or 7-port, with power supply)	Optional	Recommended if connecting GPS + TNC + backup drive simultaneously. Must be powered — unpowered hubs can cause instability.
USB Printer (e.g. Brother HL-L2350DW, HP LaserJet)	Optional	Shared across EMCOMM-NET via CUPS (installed automatically). Or connect a Wi-Fi printer directly to EMCOMM-NET.

USB Hub & Multi-Device Setup

The Pi 5 has four USB ports. A powered USB hub lets you connect GPS, TNC, backup drive, and other peripherals simultaneously. FieldComms uses udev rules to assign stable device names regardless of hub port or plug order.

DEVICE	STABLE NAME	HOW ASSIGNED	SERVICE
USB GPS receiver	/dev/gps0	udev rule matches by USB vendor/product ID	gpsd → tactical map
Digirig Mobile TNC	/dev/tnc0 /dev/digirig	udev rule matches by vendor ID + product string "Digirig Mobile". Distinguished from GPS even though both use CP2102 chip.	YAAC / Graywolf APRS
SignaLink USB TNC	/dev/tnc0 /dev/signalink	udev rule matches by Texas Instruments PCM2904 chip. No conflict with GPS.	YAAC / Graywolf APRS

LaCie / USB backup drive (labeled FIELDCOMMS)	/dev/sdX (auto-mounted)	udev backup rule triggers on label FIELDCOMMS. Drive letter assigned dynamically.	fieldcomms-backup@.service

> Both the Digirig Mobile TNC and GlobalSat BU-353-S4 GPS use the same USB chip (Silicon Labs CP2102, vendor 10c4:ea60). FieldComms distinguishes them by USB product description string. The GPS rule excludes Digirig; the TNC rule requires the Digirig product string. Verify with: udevadm info -a -n /dev/ttyUSB0 | grep -E "idVendor|idProduct|product"

```
# List all USB serial devices ls -la /dev/ttyUSB* /dev/ttyACM* 2>/dev/null # Check the stable symlinks
ls -la /dev/gps0 /dev/tnc0 2>/dev/null # Confirm GPS is outputting NMEA sentences sudo cat /dev/gps0 #
Use /dev/tnc0 in YAAC → Configure → Ports → Serial TNC
```

Complete Bill of Materials

ITEM	MODEL / SPEC	WHERE TO BUY
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**— Core Server —**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=8.5, greek=0, italic=0, link=[], rise=0, text='— Core Server —', textColor=Color(.176471,.415686,.705882,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph
Raspberry Pi 5 — 16 GB RAM	Raspberry Pi 5 Model B (16 GB)	raspberrypi.com · Adafruit · PiShop.us
Pironman MAX 5 enclosure	Pironman MAX 5 (tower, dual NVMe, OLED, fans)	52pi.com · Amazon
NVMe SSD × 2 (for RAID 1)	1 TB M.2 2280 PCIe Gen 3/4 NVMe (e.g. WD Blue SN580)	Amazon · B&H; · Newegg
Pi 5 USB-C power supply	Official Raspberry Pi 27W USB-C PD power supply	raspberrypi.com · Adafruit · Amazon
MicroSD card (boot / initial RAID setup)	32 GB Class 10 / A1 microSD	Amazon
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**— Networking —**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=8.5, greek=0, italic=0, link=[], rise=0, text='— Networking —', textColor=Color(.176471,.415686,.705882,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph
ASUS RT-BE58 Go travel router	ASUS RT-BE58 Go (Wi-Fi 7, 2.5G LAN, USB-C power)	Amazon · Best Buy · B&H;
UniFi Switch Flex 2.5G-5	Ubiquiti USW-Flex-2.5G-5 (5-port 2.5 GbE)	ui.com · Amazon · B&H;
CAT 6 Ethernet cables × 4	3 ft – 10 ft CAT 6 patch cables (router→switch, switch→Pi, spares)	Amazon · Monoprice

Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**— Radio & Comms —**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=8.5, greek=0, italic=0, link=[], rise=0, text='— Radio ', textColor=Color(.176471,.415686,.705882,1), us_lines=[]), ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=8.5, greek=0, italic=0, link=[], rise=0, text='&', textColor=Color(.176471,.415686,.705882,1), us_lines=[]), ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=8.5, greek=0, italic=0, link=[], rise=0, text=' Comms —', textColor=Color(.176471,.415686,.705882,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph
Icom IC-7300 HF transceiver	Icom IC-7300 (HF/50 MHz, built-in USB sound card)	Ham Radio Outlet · DX Engineering · Amazon
USB-A to USB-B cable (IC-7300 to laptop)	USB-A to USB-B, shielded, 6 ft	Amazon · Monoprice
Windows laptop (Winlink Express + JS8Call)	Any Windows 10/11 laptop with USB-A port and Wi-Fi	Best Buy · Amazon
Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '**— Accessories —**' 'frags': [ParaFrag(__tag__='b', bold=1, fontName='Helvetica-Bold', fontSize=8.5, greek=0, italic=0, link=[], rise=0, text='— Accessories —', textColor=Color(.176471,.415686,.705882,1), us_lines=[])] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph	Paragraph('caseSensitive': 1 'encoding': 'utf8' 'text': '' 'frags': [] 'style': 'bulletText': None 'debug': 0) #Paragraph
USB-A drive (32 GB+, labeled FIELDCOMMS)	32 GB+ USB 3.0 drive — auto-backup trigger when inserted	Amazon
External USB hard drive 1 TB+ (e.g. LaCie Rugged)	1 TB+ USB 3.0/USB-C portable — Label FIELDCOMMS	B&H; · Amazon · Best Buy
USB GPS receiver (optional)	GlobalSat BU-353-S4 or u-blox USB GPS puck	Amazon
USB TNC — Digirig Mobile (optional)	Digirig Mobile v1.x — USB soundcard + serial CAT in one unit	digirig.net · Amazon
USB TNC — SignaLink USB (optional)	Tigertronics SignaLink USB — rig-specific cable required	tigertronics.com · Ham Radio Outlet · Amazon
Powered USB hub (optional)	Anker 7-Port USB 3.0 with power adapter — powered hub required	Amazon · Best Buy
USB Printer — laser (recommended)	Brother HL-L2350DW or HP LaserJet P1102w — shared via CUPS	Best Buy · Amazon
USB Printer — portable/battery (optional)	Canon PIXMA TR150 or HP OfficeJet 200 — battery-powered	Best Buy · Amazon · Office Depot
Avery 5371 business card sheets (operator cards)	Avery 5371, 10 cards per sheet — laser or inkjet	Office Depot · Amazon · Staples

AiMesh Coverage Extension — Optional Bill of Materials

ITEM	MODEL / SPEC	WHERE TO BUY
AiMesh node router (wireless backhaul)	ASUS RT-BE58 Go (same as primary — simplest pairing) or any AiMesh-compatible ASUS router	Amazon · Best Buy · B&H;
AiMesh node router (high-capacity venues)	ASUS ZenWiFi Pro ET12 or RT-AX88U Pro — better range and capacity for large buildings	Amazon · B&H; · Costco
CAT 6 Ethernet cable (wired backhaul — recommended)	CAT 6 cable from UniFi switch to node LAN port. Length as needed.	Amazon · Monoprice · Home Depot
USB-C power bank (field power for node)	10,000+ mAh, 18W+ USB-C PD — one per node for field deployment without shore power	Amazon

DEPLOYMENT SCENARIO	RECOMMENDED SETUP
Single room EOC (≤ 2,500 sq ft)	1× RT-BE58 Go primary only — no extension needed
Multi-room EOC or large shelter (2,500–7,500 sq ft)	1× RT-BE58 Go primary + 1 AiMesh node (wireless or wired backhaul)
Large building or campus (7,500–20,000 sq ft)	1× primary + 2–3 nodes, wired backhaul recommended for all nodes
Outdoor SAR staging area	1× primary at command post + RT-BE58 Go nodes at field positions on battery

■ You do not need to pre-purchase AiMesh nodes for every deployment. Start with the primary RT-BE58 Go and add nodes only when coverage is insufficient. Pairing a new node takes under 5 minutes on site.

SD Card / Storage Sizing Guide

TIER	WHAT FITS / REQUIREMENT
Minimum (boot only)	8 GB — OS only, RAID boot loader. FieldComms runs from NVMe RAID.
Recommended	32 GB — OS + swap + logs + temp files during initial RAID setup.
Kiwix Tier 1	~3 GB additional on RAID — WikiMed + Wikipedia Mini + Wikivoyage.
Kiwix Tier 2	~10 GB additional on RAID — adds iFixit, Wikibooks, Wikiversity.
FCC Database	~600 MB on RAID — full US amateur license database (~800K records).

3. Before You Begin — Prerequisites

Complete these prerequisites before running the installer. The installer requires an internet connection for package downloads, the FCC database, and Kiwix ZIM files. After installation, the server operates fully offline.

3.1 Flash the Operating System

OS EDITION	PROS	CONS
Raspberry Pi OS Lite (64-bit) — Recommended for production	Minimal overhead, faster boot, all RAM available to FieldComms services, smaller attack surface	No local browser or GUI — all interaction via SSH or EMCOMM-NET browser
Raspberry Pi OS Desktop (64-bit) — Good for setup and troubleshooting	GUI for initial setup, Raspberry Pi Configuration tool, local browser for testing	More RAM used by desktop environment; install takes longer
Ubuntu Server 24.04 LTS — Also supported	LTS kernel, familiar to Linux admins, good for organizations standardized on Ubuntu	Slightly larger memory footprint; some package names differ

3.2 First Boot — Initial Setup

STEP	ACTION
1	On first boot, complete the initial setup wizard: set username to **fieldcomms**, set a strong password, configure locale and keyboard, expand filesystem.
2	If using Desktop edition: complete the initial desktop setup wizard.
3	Enable SSH (Lite only): run `sudo raspi-config → Interface Options → SSH → Enable`

Run a full system update before installing FieldComms (works the same on both editions — Terminal or SSH):

```
# Update all packages to latest versions sudo apt-get update && sudo apt-get full-upgrade -y # Reboot to apply kernel updates sudo reboot
```

3.3 Verify Disk Space

```
# Check available space — should show 20GB+ free for Tier 2 install df -h / # Verify RAM free -h
```

3.4 Internet Connection

Connect the Pi to the internet via Ethernet before running the installer. Wi-Fi can be used, but Ethernet is preferred for downloading large files (FCC database ~600 MB, Kiwix ZIMs up to 25 GB).

> ■ Wi-Fi is provided by the ASUS RT-BE58 Go router — the Pi does NOT run hostapd. During installation, connect the Pi to the internet via Ethernet through the UniFi switch and ASUS router. After installation, the Pi is always reachable at 192.168.50.1 via wired Ethernet.

1 Network Hardware Setup — ASUS Router + UniFi Switch

4. Step 1: Network Hardware Setup

Complete this step before running the FieldComms installer. The ASUS router must be configured with the correct LAN subnet (192.168.50.x) before FieldComms is reachable at http://192.168.50.1.

1.1 Physical Wiring

```
ASUS RT-BE58 Go WAN port → Ethernet uplink (EOC internet) or leave unplugged for offline ASUS RT-BE58 Go
LAN port → UniFi Switch Flex 2.5G-5, Port 1 (uplink) UniFi Switch Port 2 → Raspberry Pi 5 Ethernet
UniFi Switch Port 3 → Windows laptop (optional, or use Wi-Fi) UniFi Switch Ports 4–5 → Spare (second
TNC, Starlink, etc.) Power: ASUS router — USB-C 18W PD adapter or power bank UniFi switch — included
USB-C 5V/1A adapter (or PoE from uplink) Pi 5 — 27W USB-C PD supply via Pironman MAX 5 enclosure PSU
```

1.2 Configure the ASUS RT-BE58 Go

STEP	ACTION
1	Power on the router. Connect a device to it via Wi-Fi (default SSID on label) or Ethernet. Open http://192.168.50.1 (or the default router IP on the label).
2	Complete the router setup wizard. When it asks for WAN, choose Automatic IP (DHCP) or skip if no internet is connected.
3	Go to **LAN** → **LAN IP**. Set the LAN IP to **192.168.50.1**, subnet **255.255.255.0**. Apply.
4	Go to **LAN** → **DHCP Server**. Set IP pool: **192.168.50.100 – 192.168.50.200**. Apply.
5	Change the ASUS router admin password to something strong. Record it on a piece of paper stored on the FIELDCOMMS USB drive.
6	Set Manually Assigned IP for the Pi: go to **LAN** → **DHCP Server** → **Manually Assigned IP**. Enter the Pi's MAC address → assign IP **192.168.50.1**. (Or set the Pi's static IP manually in Step 4.)
7	Set the Wi-Fi SSID: go to **Wireless** → **General**. Set both the 2.4 GHz and 5 GHz SSIDs to **EMCOMM-NET**. Set a strong WPA3/WPA2 password. Apply.

WAN SOURCE	SETTING	WHEN TO USE
No WAN (offline only)	Leave WAN unplugged	Default field posture — all FieldComms tools work without internet
Ethernet uplink	WAN → Automatic IP	EOC with site network — enables NWS alerts, HF prop data, APRS-IS
USB smartphone tether	Enable tethering on phone, plug into ASUS USB port, WAN → USB	Field site with cellular coverage
WISP (connect to site Wi-Fi)	ASUS Web UI → Operation Mode → WISP	Hospital / shelter with guest Wi-Fi available

1.3 AiMesh Coverage Extension (Optional)

STEP	ACTION
1	Factory reset the node router. Hold reset button for 5–10 seconds until power LED flashes.
2	Power on the node within 30 ft of the primary router for initial pairing.

3	On primary router (http://192.168.50.1): go to **AiMesh** → **Add AiMesh Node**.
4	Select the new node. Click Connect. The primary pushes EMCOMM-NET config to the node (1–3 minutes).
5	Move the node to its final position. Test coverage with a phone on EMCOMM-NET.

2 Download & Run the FieldComms Installer

5. Step 2: Download & Run the FieldComms Installer

Method A — Direct download to the Pi (requires internet on Pi)

Lite: SSH into the Pi and run these commands. **Desktop:** open a Terminal window (taskbar → Terminal) and run the same commands.

```
# Create working directory mkdir -p ~/fieldcomms-install && cd ~/fieldcomms-install # Download the FieldComms package from GitHub or your storage location wget -O fieldcomms-v1.zip https://github.com/KE4CON/CrossPlatformAPRS/releases/latest/download/fieldcomms-v1.zip # Unzip the package unzip fieldcomms-v1.zip cd fieldcomms
```

Method B — Copy from your computer using SCP (Lite/headless)

```
# Run this on YOUR COMPUTER, not the Pi scp fieldcomms-v1.zip fieldcomms@[pi-ip-address]:~/ # Then SSH into the Pi and unzip ssh fieldcomms@[pi-ip-address] mkdir -p ~/fieldcomms-install mv fieldcomms-v1.zip ~/fieldcomms-install/ cd ~/fieldcomms-install && unzip fieldcomms-v1.zip cd fieldcomms
```

Method B2 — Download directly on the Pi Desktop (Desktop edition)

OPTION	HOW
Browser download	Open Chromium on the Pi. Go to the GitHub releases page or download URL. Download fieldcomms-v1.zip — it saves to ~/Downloads/ automatically.
USB drive	Copy fieldcomms-v1.zip to a USB drive. Insert the USB into the Pi. Open File Manager, drag the zip to your home folder.

```
# If downloaded to ~/Downloads/: mkdir -p ~/fieldcomms-install cp ~/Downloads/fieldcomms-v1.zip ~/fieldcomms-install/ # Unzip (same for all methods): cd ~/fieldcomms-install && unzip fieldcomms-v1.zip cd fieldcomms
```

Method C — USB drive transfer (fully offline)

Lite: commands below via SSH. **Desktop:** use File Manager to copy the zip from USB to your home folder, then open a Terminal for the unzip step.

```
# Copy fieldcomms-v1.zip to a USB drive on your computer # Insert USB drive into the Pi, then: sudo mount /dev/sda1 /mnt # or check lsblk for your device cp /mnt/fieldcomms-v1.zip ~/fieldcomms-install/ cd ~/fieldcomms-install && unzip fieldcomms-v1.zip cd fieldcomms
```

Launch the Installer

Lite: run via SSH. **Desktop:** open a Terminal window from the taskbar or Accessories → Terminal. The installer runs identically on both.

```
cd ~/fieldcomms-install/fieldcomms sudo bash scripts/install.sh
```

3 Installer Configuration Options

6. Step 3: Installer Configuration Options

PROMPT	DEFAULT	DESCRIPTION
Station callsign	W8ABC	Your amateur callsign. Patched into all HTML pages.
Station latitude	42.3153	Decimal degrees. Used for APRS station marker, NWS alerts, propagation, distance calc.
Station longitude	-88.4473	Decimal degrees (negative = West). Default: Woodstock IL (McHenry County seat).
Wi-Fi AP SSID	EMCOMM-NET	The Wi-Fi network name that field devices connect to.
Wi-Fi AP password	fieldcomms2026	WPA2 password for EMCOMM-NET. Change this for operational security.
Server IP address	192.168.50.1	The Pi's static IP on EMCOMM-NET. Devices browse to http://192.168.50.1/
Download FCC database	N	~600 MB download of the FCC amateur license database. Recommended: Y
Kiwix tier	1	Tier 1: WikiMed + Wikipedia Mini + Wikivoyage (~2.5 GB). Tier 2 adds iFixit (~5 GB more).

What the Installer Does Automatically

- Installs system packages: Python 3, nginx, kiwix-tools, rsync, gpsd, ufw, mdadm, java
- Creates the **fieldcomms** system user and directory structure at /opt/fieldcomms/
- Creates a Python virtual environment and installs Flask, flask-cors, requests, gpsd-py3
- Deploys all 30 HTML pages to /opt/fieldcomms/html/ and patches your callsign and coordinates into each file
- Installs all 14 systemd service and timer units and enables them at boot
- Configures nginx with the correct web root (/opt/fieldcomms/html/) and proxy rules for all API ports
- **Does NOT configure a Wi-Fi access point** — Wi-Fi is handled by the ASUS RT-BE58 Go router. The Pi connects as a wired client only.
- Configures the Pi static IP (192.168.50.1/24) on the Ethernet interface (eth0) via NetworkManager
- Configures the firewall (ufw) — opens all required ports
- Downloads and installs YAAC (Java APRS client) and Graywolf to /opt/yaac/ and /opt/graywolf/
- Installs the USB backup udev rule (plug in a USB drive labeled FIELDCOMMS to trigger auto-backup)
- Downloads and installs Pat Winlink (browser-based Winlink backup client, port 8090)
- Runs the Kiwix ZIM downloader for the selected content tier
- Downloads the FEMA ICS forms PDFs to /opt/fieldcomms/data/ics_forms/ (22 forms)
- Optionally downloads and builds the FCC amateur license database (~600 MB)
- Starts all services and verifies they are running

■ The ASUS RT-BE58 Go router must be configured before running the installer — see Step 1. The Pi does NOT run hostapd or dnsmasq. All Wi-Fi, DHCP, and SSID management is handled by the ASUS router.

4 Raspberry Pi 5 — Static IP Configuration

7. Step 4: Raspberry Pi 5 — Static IP Configuration

The Pi must always be reachable at 192.168.50.1. If the installer already set this via a DHCP reservation in the ASUS router, verify with `ip addr show eth0` and skip to Step 5 if 192.168.50.1/24 is shown.

Path A — Raspberry Pi OS Lite (SSH)

```
# SSH into the Pi (use the IP shown in your router DHCP table) ssh fieldcomms@[current-pi-ip] # Find the Ethernet connection name nmcli con show # Usually: "Wired connection 1" or "Preconfigured Connection 1" # Set static IP sudo nmcli con mod "Wired connection 1" \ ipv4.addresses 192.168.50.1/24 \ ipv4.method manual # Apply the change sudo nmcli con up "Wired connection 1" # Verify — should show inet 192.168.50.1/24 ip addr show eth0
```

Path B — Raspberry Pi OS Desktop (GUI)

STEP	ACTION
1	Click the **network icon** in the taskbar (top-right corner).
2	Click **Advanced Options** → **Edit Connections**.
3	Select your **Wired connection** and click the gear/edit icon.
4	Click the **IPv4 Settings** tab.
5	Change **Method** from "Automatic (DHCP)" to **Manual**.
6	Click **Add** and enter: Address: 192.168.50.1 · Netmask: 255.255.255.0 · Gateway: (leave blank)
7	Click **Save**, then close the Network Connections window.
8	Disconnect and reconnect the Ethernet cable, or reboot the Pi.
9	Open the **Terminal** and verify: `ip addr show eth0` — should show **inet 192.168.50.1/24**.
10	Open **Chromium** on the Pi and go to http://192.168.50.1 to confirm FieldComms loads locally.

■ Alternatively, open a Terminal on the desktop and use the same nmcli commands shown in Path A. The terminal method works identically on the Desktop edition.

Path C — raspi-config (if available)

```
# Open raspi-config: sudo raspi-config # Navigate to: System Options → Network → (set static IP if option is present) # NOTE: Static IP configuration via raspi-config was removed in Raspberry Pi OS Bookworm. # If the option is not visible, use Path A (nmcli) or Path B (GUI Network Connections).
```

4b RAID 1 NVMe Setup — Pironman MAX 5

7b. RAID 1 NVMe Setup — Pironman MAX 5

The Pironman MAX 5 enclosure has two M.2 NVMe slots. Configure them as a RAID 1 mirror so both drives contain identical data. If one drive fails, the system continues running on the surviving drive with no data loss and no operator action required.

> Complete this step before running the FieldComms installer. The RAID array must be assembled and the OS installed on it before FieldComms is deployed. If your Pi OS is already on a single drive, back it up first.

Physical Installation

STEP	ACTION
1	Install both 1 TB NVMe SSDs into the Pironman MAX 5 M.2 slots. Slot 1 (primary) → /dev/nvme0n1 · Slot 2 (mirror) → /dev/nvme1n1
2	Boot from a Raspberry Pi OS microSD card for the initial RAID setup. After RAID is built and OS is installed, the SD card is removed.

Build the RAID 1 Array

```
# Install mdadm (RAID management tool) sudo apt-get install -y mdadm # Create RAID 1 array from the two
NVMe drives sudo mdadm --create --verbose /dev/md0 --level=1 --raid-devices=2 \ /dev/nvme0n1
/dev/nvme1n1 # Confirm array is building (will show [=>..............] resync progress) cat
/proc/mdstat # Create a filesystem on the RAID array sudo mkfs.ext4 -L fieldcomms-raid /dev/md0 # Mount
it sudo mkdir -p /mnt/raid sudo mount /dev/md0 /mnt/raid # Copy running OS to RAID array sudo apt-get
install -y rsync sudo rsync -axv --progress / /mnt/raid/ # Save RAID configuration so it persists across
reboots sudo mdadm --detail --scan | sudo tee -a /mnt/raid/etc/mdadm/mdadm.conf
```

RAID Health Reference

SITUATION	WHAT TO DO
Both drives show [UU]	Normal healthy state. No action needed.
One drive fails — [_U] or [U_]	1. Power down. 2. Remove failed drive. 3. Insert new 1 TB NVMe. 4. Power on. 5. Run: sudo mdadm /dev/md0 --add /dev/nvme1n1. 6. Array rebuilds automatically (~30 min).
Check RAID health anytime	cat /proc/mdstat or sudo mdadm --detail /dev/md0

> RAID 1 protects against drive failure — it is NOT a backup. Both drives contain identical data, so accidental deletion or corruption affects both drives simultaneously. Continue to use the USB auto-backup (insert FIELDCOMMS USB drive) for regular data backups.

5 Kiwix Offline Library Setup

8. Step 5: Kiwix Offline Library Setup

Kiwix provides an offline web library served on port 8081. All devices on EMCOMM-NET can browse it at http://192.168.50.1:8081. Content is packaged as ZIM files.

TIER	NAME	SIZE	DESCRIPTION
1 — Essential	WikiMed — Medical Encyclopedia	~471 MB	Offline medical encyclopedia: symptoms, treatments, drugs, procedures. Critical for mass-casualty events.
1 — Essential	Wikipedia (Mini)	~1.2 GB	Compact English Wikipedia — essential facts, geography, science.
1 — Essential	Wikivoyage	~820 MB	Emergency shelters, evacuation routes, local resources and travel logistics.
2 — Extended	Wikibooks — How-To & Field Manuals	~4.2 GB	First aid, survival, ham radio, electronics, cooking, field skills.
2 — Extended	iFixit — Equipment Repair Manuals	~3.1 GB	Step-by-step repair guides for electronics, tools, vehicles.
2 — Extended	Wikipedia (Full English)	~22 GB	Complete English Wikipedia. Large download — requires 30 GB+ free.

```
# Check what is installed and service status sudo bash /opt/fieldcomms/scripts/kiwix_setup.sh --status # Add Tier 2 content later (resumes interrupted downloads) sudo bash /opt/fieldcomms/scripts/kiwix_setup.sh --tier 2 # Service management sudo systemctl status kiwix sudo systemctl restart kiwix # after adding new ZIM files
```

■ ZIM downloads are resumable. If a download is interrupted, simply re-run kiwix_setup.sh with the same --tier flag. curl will resume from where it stopped.

6 FCC Amateur Database

9. Step 6: FCC Amateur Database

The FCC database provides offline callsign lookup for the entire US amateur license database (~800,000 licensees). Once built, lookups are instant and completely offline.

```
# If you selected "y" during install, it already ran. # To build or rebuild manually: sudo -u fieldcomms \ /opt/fieldcomms/venv/bin/python \ /opt/fieldcomms/python/build_fcc_db.py # Automatic weekly refresh timer: sudo systemctl status fcc-refresh.timer # Force immediate refresh: sudo systemctl start fcc-refresh.service
```

7 APRS Setup — Graywolf + YAAC

10. Step 7: APRS Setup (Graywolf + YAAC)

FieldComms supports two simultaneous APRS clients: Graywolf TNC (port 8080) and YAAC — Yet Another APRS Client (port 8082). Both are optional — FieldComms runs fully without them.

> ■ The FieldComms installer attempts to download and install both YAAC and Graywolf automatically. If internet was available during installation, they should already be present. Skip to Section 9.3 (YAAC port configuration) if the installer succeeded.

9.1 Verify Automatic Installation

```
# Check if Java is installed java -version # Check if YAAC was installed ls -lh /opt/yaac/YAAC.jar # Check if Graywolf was installed ls -lh /opt/graywolf/graywolf.jar # Check service status sudo systemctl status yaac sudo systemctl status graywolf
```

9.2 Manual Install — If Automatic Install Failed

```
# Step 1 — Install Java runtime (required by both clients) sudo apt install -y default-jre # Step 2 — Install YAAC sudo mkdir -p /opt/yaac cd /tmp && wget http://www.ka2ddo.org/ka2ddo/YAAC.zip sudo unzip YAAC.zip "*.jar" -d /opt/yaac/ sudo mv /opt/yaac/YAAC*.jar /opt/yaac/YAAC.jar 2>/dev/null || true sudo chown -R fieldcomms:fieldcomms /opt/yaac # Step 3 — Install Graywolf sudo mkdir -p /opt/graywolf sudo wget -O /opt/graywolf/graywolf.jar \
https://github.com/vk2tds/graywolf/releases/latest/download/graywolf.jar sudo chown -R fieldcomms:fieldcomms /opt/graywolf # Step 4 — Enable services sudo systemctl enable --now yaac sudo systemctl enable --now graywolf
```

9.3 YAAC Port Configuration (Required — First Run)

STEP	ACTION
1	Run YAAC once on a desktop/monitor: `java -jar /opt/yaac/YAAC.jar`
2	Go to **File** → **Configure** → **Web Server** tab.
3	Set **Port: 8082** · Check **Enable REST API** · Check **Enable WebSocket** · Click Save.
4	Close YAAC. The yaac.service will use these settings when running headlessly.

9.4 Connect a TNC or Radio Interface

INTERFACE TYPE	PORT TYPE IN YAAC	SETTINGS
USB TNC (Digirig, SignaLink, etc.)	Serial TNC	/dev/tnc0, baud rate 9600 or 1200
KISS TNC over TCP (Direwolf)	KISS TNC (TCP)	Host: localhost, Port: 8001
AGW packet engine (Direwolf)	AGW	Host: localhost, Port: 8000
APRS-IS internet (if WAN connected)	APRS-IS	Server: noam.aprs2.net:14580, Filter: r/42.31/-88.44/50

9.5 Verify APRS is Working

```
# Test YAAC REST API (should return JSON list of heard stations) curl http://localhost:8082/api/stations
# Test Graywolf REST API curl http://localhost:8080/api/stations # Check service logs if no stations
appear journalctl -u yaac -n 30 journalctl -u graywolf -n 30
```

8 Printer Setup — Three Connection Options

11. Step 8: Printer Setup

FieldComms has print buttons on 17 pages. All printing uses the browser standard window.print() function, which sends the job to whatever printer the operator's browser can reach. Three printer connection methods are supported:

Option A — Client Device Prints to Its Own Printer (Simplest)

Each operator's laptop or tablet prints to its own locally connected printer. No Pi configuration needed. Zero setup required — works immediately.

■ This is the simplest option and requires zero configuration on the Pi. The only requirement is that the operator's device has a printer configured.

Option B — USB Printer Shared via Pi (CUPS)

A USB printer is connected directly to the Pi. CUPS shares it across EMCOMM-NET. Installed automatically by the FieldComms installer. CUPS admin UI at http://192.168.50.1:631.

B.2 Add the USB Printer via CUPS Web Admin

STEP	ACTION
1	Plug the USB printer into one of the Pi's USB ports (or powered USB hub).
2	On any device on EMCOMM-NET, open a browser and go to: **http://192.168.50.1:631**
3	Click **Administration** → **Add Printer**.
4	If prompted for credentials, enter the Pi username and password (e.g. fieldcomms / your password).
5	Your USB printer appears in the **Local Printers** list. Select it and click **Continue**.
6	Enter a Name (e.g. FieldComms-Printer), Description, and Location. Check **Share This Printer**. Click Continue.
7	Select the correct printer driver. Click **Add Printer**.
8	Set default options (paper size: Letter) and click **Set Default Options**.
9	Click **Print Test Page** to confirm the printer is working.

B.3 Add the Shared Printer on Operator Devices

DEVICE TYPE	HOW TO ADD THE SHARED PRINTER
Windows laptop	Settings → Bluetooth & devices → Printers & scanners → Add a printer. Windows discovers the CUPS printer automatically on EMCOMM-NET.
Mac / macOS	System Settings → Printers & Scanners → + Add Printer. The printer appears under the Default tab via Bonjour. Select it and click Add.
iPad / iPhone	iOS prints via AirPrint. CUPS with Avahi shares the printer as AirPrint-compatible. Tap Share → Print in any FieldComms page and select the printer.
Android	Install the CUPS Print app (Google Play). Add printer at IP 192.168.50.1, port 631, protocol IPP.
Chromebook	Settings → Advanced → Printing → Add Printer. Enter: Address=192.168.50.1, Protocol=IPP, Queue=/printers/FieldComms-Printer.

B.4 Recommended Printers for Field Use

PRINTER	TYPE	WHY IT WORKS WELL
Brother HL-L2350DW	Laser, USB + Wi-Fi	Excellent Linux support. Fast, compact, duplex, great field durability.
HP LaserJet P1102w	Laser, USB + Wi-Fi	Excellent Linux driver via HPLIP. Fast, reliable, low cost per page.
Canon PIXMA TR150	Inkjet, USB + Wi-Fi, battery	Portable battery-powered option for mobile deployments. ~200 pages per charge.
HP OfficeJet 200	Inkjet, USB + Wi-Fi, battery	Portable battery printer with larger paper tray. Good field option without shore power.

Option C — Network Printer on EMCOMM-NET (Simplest Shared Setup)

A Wi-Fi or Ethernet printer is connected directly to the ASUS router or UniFi switch — no Pi involvement at all. All devices on EMCOMM-NET can discover and print to it.

CONNECTION METHOD	HOW TO SET UP
Wi-Fi (wireless)	Use the printer's own control panel to connect it to EMCOMM-NET. Enter the password when prompted. The printer receives an IP in the 192.168.50.100–200 range from the ASUS router DHCP server.
Ethernet (wired)	Run a CAT 6 cable from the printer's Ethernet port to any spare port on the UniFi Switch (Ports 3–5). The printer receives an IP automatically. Check the ASUS router DHCP table for the assigned IP.

Reserve a Static IP for the Printer (Recommended)

```
# On the ASUS router web UI (http://192.168.50.1): # LAN → DHCP Server → Manually Assigned IP # Enter the printer MAC address → assign a fixed IP, e.g.: 192.168.50.10 # Reboot the printer to pick up the reservation # Confirm: ping 192.168.50.10
```

Printer Option Comparison

	Option A — Client Printer	Option B — USB via Pi / CUPS	Option C — Network Printer
Pi configuration needed	None	CUPS (auto-installed)	None
One printer for all operators	Each device needs own printer	Shared across EMCOMM-NET	Shared across EMCOMM-NET
Works without internet	Yes	Yes	Yes
iOS / AirPrint support	Requires AirPrint printer	Yes via CUPS + Avahi	Yes if AirPrint printer
Battery printer support	Yes	Yes (USB connected)	Yes (Wi-Fi connected)
Best for	Single operator or tablet with own printer	Any USB printer, shared to all devices	Wi-Fi/Ethernet printer already on site

■ For most K9ESV field activations, Option B (USB printer via CUPS) is recommended. A Brother HL-L2350DW or HP LaserJet plugged into the Pi USB hub covers all ICS forms, net logs, and cheat sheets needed for a typical activation.

9 Pat Winlink — Verify & Configure

12. Step 9: Pat Winlink — Verify & Configure

Pat Winlink is installed automatically by the FieldComms installer. It runs as pat.service and provides a browser-based backup Winlink client at http://192.168.50.1:8090.

```
# Check pat.service status sudo systemctl status pat # If not running: sudo systemctl start pat && sudo systemctl enable pat # Confirm port 8090 is listening ss -tlnp | grep 8090 # Add your Winlink password: sudo nano /opt/fieldcomms/.config/pat/config.json # Verify / update these fields: # "mycall": "K9ESV" ← set by installer # "secure_login_password": "xxxxx" ← add your Winlink password # "http_addr": "0.0.0.0:8090" ← must be 0.0.0.0 for EMCOMM-NET access
```

10 Windows Laptop — Winlink Express + JS8Call

12. Step 10: Windows Laptop Setup

The Windows laptop is the primary HF digital station. It runs Winlink Express and JS8Call with the IC-7300 connected via a single USB cable. Connect the laptop to EMCOMM-NET (Wi-Fi) or the UniFi switch Port 3 (wired Ethernet).

9.1 IC-7300 One-Time Radio Setup

IC-7300 MENU PATH	SETTING
SET → Connectors → CI-V Baud Rate	115200
SET → Connectors → CI-V Transceive	ON
SET → Connectors → USB Send/Keying → USB Send	RTS
SET → Connectors → MOD Input → USB MOD Level	40–50%
COMP button (speech compression)	OFF
Mode for digital operation	USB-D

9.2 Install Winlink Express + VARA HF

STEP	ACTION
1	Download Winlink Express from: **winlink.org/client-software**. Run installer, accept defaults.
2	On first launch: enter your callsign, Winlink password, and grid square EN52 (McHenry County, IL).
3	Settings → Radio Setup: Radio=IC-7300, Control Port=(check Device Manager), Baud Rate=115200, PTT via CAT=checked.
4	Settings → Sound Card: Input=USB Audio CODEC (IC-7300), Output=USB Audio CODEC (IC-7300).
5	Download and install VARA HF from: winlink.org → VARA → VARA HF Modem. Set same audio devices in VARA HF Settings.
6	Test: Open a VARA HF session in Winlink Express → connect to any Winlink RMS gateway on 40m to confirm audio and CAT control work.

9.3 Install JS8Call

STEP	ACTION
1	Download JS8Call from: **js8call.com** → Windows installer. Run installer, accept defaults.
2	Launch JS8Call. Open File → Settings (F2). General tab: My Call=K9ESV, My Grid=EN52.
3	Audio tab: Input=USB Audio CODEC (IC-7300), Output=USB Audio CODEC (IC-7300).
4	Radio tab: Rig=IC-7300, PTT Method=CAT, Serial Port=(same COM port as Winlink), Baud Rate=115200.
5	Reporting tab (CRITICAL): TCP Server Hostname=**0.0.0.0** (MUST change from 127.0.0.1). Enable TCP Server API=checked. TCP Server Port=**2442**. Accept TCP Requests=checked.
6	Click OK. Restart JS8Call for API settings to take effect.

7	Set VFO to 7.078 MHz (JS8 40m calling frequency). Mode → Normal.

9.4 Windows Firewall — Allow JS8Call Port 2442

```
1. Search "Windows Defender Firewall" → Advanced Settings 2. Inbound Rules → New Rule → Port → TCP → Specific port: 2442 → Allow → All profiles 3. Name the rule: JS8Call API
```

9.5 Find the Windows Laptop IP on EMCOMM-NET

```
# Run in Windows Command Prompt: ipconfig # Note the IPv4 Address for the EMCOMM-NET adapter (e.g. 192.168.50.105) # Set a DHCP reservation (recommended — gives laptop a fixed IP): # ASUS Web UI → LAN → DHCP Server → Manually Assigned IP # Enter laptop MAC address → assign 192.168.50.2 # Configure the FieldComms dashboard JS8Call card: # 1. Connect to EMCOMM-NET, open http://192.168.50.1 # 2. Find the JS8Call card (purple) in Amateur Radio section # 3. Tap/click the card # 4. Enter the Windows laptop IP (e.g. 192.168.50.105) # 5. Card saves IP and opens http://192.168.50.105:2442
```

SWITCHING TO...	PROCEDURE
Winlink Express	Close/disconnect JS8Call → open VARA HF → it reclaims USB audio → open Winlink Express
JS8Call	Finish any Winlink TX → close VARA HF → minimize Winlink Express → open JS8Call and click Connect
Both simultaneously (two radios)	Connect a second HF radio to the laptop via a separate USB audio and CAT interface. Configure JS8Call to use the second radio port. Both can run at the same time with no conflicts.

1
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First Boot Verification

13. Step 11: First Boot Verification

```
# Check all FieldComms services for svc in fcc-lookup health-monitor deadmans ics-platform kiwix nginx
pat; do echo "$svc: $(systemctl is-active $svc)" done # Test the dashboard from the Pi itself curl -s
http://localhost/ | head -5 # Test API endpoints curl http://localhost:5050/health # FCC API health
check curl http://localhost:5051/health # System health monitor # Verify Pi has the correct static IP ip
addr show eth0 # Should show: inet 192.168.50.1/24
```

14. Network Architecture Reference

Network Topology

Wi-Fi is provided by the ASUS RT-BE58 Go router. The Pi is a wired client with a static IP of 192.168.50.1, connected via the UniFi Switch Flex 2.5G-5. The Pi does not run hostapd or dnsmasq.

DEVICE	IP ADDRESS	ROLE	CONNECTION
ASUS RT-BE58 Go	192.168.50.1 (LAN gateway)	Wi-Fi AP + DHCP server	WAN: site Ethernet or USB tether (optional)
UniFi Switch Flex 2.5G-5	N/A (Layer 2 switch)	Wired distribution	Port 1 → ASUS router · Port 2 → Pi · Ports 3–5 spare
Raspberry Pi 5	192.168.50.1 (static)	FieldComms application server	Wired Ethernet via UniFi switch
Windows laptop	192.168.50.2 (recommended static)	Winlink Express + JS8Call station	Wi-Fi (EMCOMM-NET) or wired via UniFi Port 3
Field devices (phones, tablets)	192.168.50.100–200 (DHCP)	Operator browsers	Wi-Fi (EMCOMM-NET, 2.4 GHz or 5 GHz)

■ Recommended: set the ASUS router LAN IP to 192.168.50.254 and the Pi static IP to 192.168.50.1. This keeps the FieldComms URL clean and the router admin clearly separate at http://192.168.50.254.

13. Web Dashboard Reference

PAGE	URL	DESCRIPTION
Dashboard	/	Main hub — UTC clock, NWS weather alerts, APRS table, tool cards, DMS status
Amateur Net Control	/netcontrol.html	Multi-net logger, FCC autofill, precedence, ICS-309 export, server sync
Starcom Net Logger	/starcom.html	Public safety net log with Radio IDs, sc- prefix nets, ICS-309 export
Net Observer	/observer.html?net=NETNAME	Read-only net viewer — 15-second auto-refresh, no login
Member Roster	/roster.html	Directory with 11 certs, 13 equipment fields, 4 roles, CSV import/export
Resource Board	/resources.html	Equipment, vehicle, personnel tracking with status cycling
Tactical APRS Map	/tactical.html	Leaflet map — Graywolf + YAAC merged, live WebSocket, APRS symbols
Starcom Resource Map	/resmap.html	Unit positioning map with zone and polygon drawing tools
Callsign Lookup	/callsign.html	FCC local database — ~800K licensees, instant offline search
ICS Platform	/ics/	Command, Operations, Planning, Logistics, Finance sections
ICS-213	/ics213.html	General Message form with print output and form log
ICS-214	/ics214.html	Activity Log with personnel and timestamped activity rows
ICS-309	/ics309.html	Communications Log with incident archiving
NTS Radiogram	/nts.html	ARRL-format radiogram generator and traffic log
Winlink Form Import	/winlink-import.html	Import Winlink Express XML forms to incident archive
Pre-Flight	/preflight.html	GO/CAUTION/NO-GO deployment readiness assessment
Health Monitor	:5051/health	Live CPU/memory/disk/temp, all service status, GPS, internet
Pat Winlink	:8090	Browser-based backup Winlink client
Kiwix Library	:8081/	Offline reference library — WikiMed, Wikipedia, iFixit
Print Center	/printcenter.html	All printable documents in one place + incident cover sheet generator

14. Service & Port Reference

PORT	SERVICE	SYSTEMD UNIT	DESCRIPTION
80	nginx	nginx.service	Web frontend — serves all HTML pages
5050	FCC Lookup Server	fcc-lookup.service	Main API: callsign lookup, net logs, roster, resources, DMS, ICS forms
5051	Health Monitor	health-monitor.service	System health: CPU temp, memory, disk, GPS, all service status
5055	ICS Platform	ics-platform.service	ICS incident management: incidents, objectives, resources, T-cards
5056	References API	fieldcomms-refs.service	Reference library API
8080	Graywolf APRS	(graywolf.service)	APRS TNC client — REST API + WebSocket live stream
8081	Kiwix Library	kiwix.service	Offline web library — WikiMed, Wikipedia, iFixit, etc.
8082	YAAC APRS	yaac.service	YAAC Java APRS client — REST API + WebSocket
8083	Tile Server	fieldcomms-tiles.service	Offline map tile server (MBTiles)
8090	Pat Winlink	pat.service	Winlink email-over-radio — Packet, VARA HF, VARA FM
631	CUPS Printer	cups.service	Print server — USB printer shared to EMCOMM-NET
2442	JS8Call	(JS8Call app on laptop)	HF digital keyboard messaging — TCP API on Windows laptop

15. Maintenance & Updates

```
# Interactive maintenance menu: sudo bash /opt/fieldcomms/scripts/update.sh # Menu options: # 1) Restart all services # 2) Stop all services # 3) Check service status # 4) View live logs # 5) Refresh FCC database # 6) Fetch repeater data # 7) Update web files from current directory # 8) Backup data to /tmp # 9) Show disk usage
```

USB Auto-Backup

Insert a USB drive formatted with the label **FIELDCOMMS** to trigger an automatic backup of all runtime data. The backup copies /opt/fieldcomms/data/ to a timestamped folder on the USB drive.

```
# Format a USB drive with label FIELDCOMMS (Linux): sudo mkfs.vfat -n FIELDCOMMS /dev/sda1 # Or on Windows: format as FAT32, set volume label to FIELDCOMMS # Backup destination on USB: # /media/fieldcomms/backup/YYYYMMDD_HHMMSS
```

16. Troubleshooting

SYMPTOM	LIKELY CAUSE / FIX
http://192.168.50.1 not reachable	Pi not running, or ASUS router not configured with 192.168.50.x subnet. Check Pi power LED. Verify router LAN IP is 192.168.50.1.
Dashboard loads but cards give errors	Not connected to EMCOMM-NET — device is on a different network. Check Wi-Fi — must show EMCOMM-NET.
FCC lookup returns no results	FCC database not yet built. Run: sudo systemctl start fcc-refresh.service (needs internet)
APRS map shows no stations	Graywolf or YAAC not running, or no RF received yet. Check Health Monitor for service status. Confirm antenna connected.
Winlink form import fails	Wrong file type — must be the XML attachment, not the message body. In Winlink Express, right-click .xml attachment → Save As → then import.
Service dot is red on Health Monitor	Background service has stopped. SSH to Pi: sudo systemctl restart
Pat Winlink not accessible at port 8090	pat.service not running. Run: sudo systemctl start pat && sudo systemctl enable pat
JS8Call card does not open	TCP Server API not enabled in JS8Call, or wrong IP entered. Verify: File → Settings → Reporting → Enable TCP Server API, port 2442, hostname 0.0.0.0.
Printer not visible on EMCOMM-NET	avahi-daemon not running, or CUPS printer not shared. Check: sudo systemctl status avahi-daemon cups. Verify Share This Printer is checked in CUPS UI.
Issue: Services crash on startup	Low disk space or permission error. Check: df -h / and ls -la /opt/fieldcomms/. Verify fieldcomms user owns /opt/fieldcomms/.

■ For any issue not covered here — open the Health Monitor at http://192.168.50.1:5051/health or check the system log with: journalctl -u fcc-lookup -n 50

17. Quick Reference Card

ITEM	VALUE
FieldComms dashboard	http://192.168.50.1
Wi-Fi network (SSID)	EMCOMM-NET
Pi static IP	192.168.50.1
ASUS router admin	http://192.168.50.1 (or http://192.168.50.254 if reconfigured)
CUPS printer admin	http://192.168.50.1:631
Pat Winlink (backup)	http://192.168.50.1:8090
Kiwix offline library	http://192.168.50.1:8081
Health monitor (raw)	http://192.168.50.1:5051/health
JS8Call TCP API (laptop)	http://[laptop-ip]:2442
Callsign K9ESV	McHenry County Emergency Services Volunteers and EMA
McHenry County grid	EN52
Default coordinates	42.3153 N / 88.4473 W (Woodstock, IL)
Install log	/var/log/fieldcomms-install.log
Application data	/opt/fieldcomms/data/
HTML pages	/opt/fieldcomms/html/
Update script	sudo bash /opt/fieldcomms/scripts/update.sh

Operator Workstation Options — Raspberry Pi 500 / 500+ + Raspberry Pi Monitor

FieldComms is a browser-based system — any device with a modern browser and Wi-Fi can connect to EMCOMM-NET and access every tool. The Raspberry Pi 500 paired with the official Raspberry Pi Monitor provides a purpose-built, low-cost, low-power operator workstation that is fully Pi ecosystem compatible and ideal for fixed EOC positions, shelter check-in stations, and net control operator desks. Funding through ARPA-E grants, FEMA BRIC, or 501(c)(3) equipment donations may cover these workstations.

Pi 500 vs Pi 500+ vs Pi 400 — Comparison

SPEC	Pi 400 (2020)	Pi 500 (Dec 2024) ★ Recommended	Pi 500+ (Sept 2025)
Processor	Cortex-A72 @ 1.8 GHz (Pi 4 generation)	**Cortex-A76 @ 2.4 GHz (Pi 5 generation)**	Cortex-A76 @ 2.4 GHz (Pi 5 generation)
RAM	4 GB DDR4	**8 GB LPDDR4X**	16 GB LPDDR4X
Storage	microSD only	microSD only (32 GB A2 included)	256 GB NVMe built-in + microSD slot
Keyboard	Membrane	Membrane	Mechanical (Gateron Blue KS-33)
Wi-Fi	802.11ac dual-band	802.11ac dual-band	802.11ac dual-band
Ethernet	Gigabit	Gigabit	Gigabit
USB ports	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0	2× USB 3.0, 1× USB 2.0
Display output	2× micro-HDMI	2× micro-HDMI	2× micro-HDMI
Production until	Jan 2028	**TBD (current model)**	TBD (current model)
Best for FieldComms?	Acceptable — older chip, only 4 GB RAM	**✓ YES — Pi 5 chip, 8 GB RAM, current generation**	Overkill for browser use Justified only if also running heavy Linux apps locally

Raspberry Pi Monitor — 15.6"

SPEC	DETAIL
Panel	15.6" IPS LCD, 1920×1080 Full HD @ 60 Hz
Video input	Standard HDMI 1.4 (full-size) — requires micro-HDMI to HDMI cable with Pi 500
Audio	2× front-facing stereo speakers, 3.5mm headphone jack
Power	USB-C, 5V/1.5A — can power from Pi 500 USB port at 60% brightness/volume, or from separate USB-C supply for 100% brightness/volume
Mounting	Integrated angle-adjustable kickstand, VESA mount, wall hanger
Dimensions	Slim 15.6" form — similar footprint to a laptop
Colors	Red/White (matching Pi 500 keyboard) or Black
Where to buy	raspberrypi.com · Adafruit · PiShop.us · GigaParts · Amazon

■ The Pi 500 has micro-HDMI ports; the Raspberry Pi Monitor has a standard full-size HDMI input. You need a micro-HDMI to standard HDMI cable (or adapter). The Pi 500 desktop kit ($120) includes one. When ordering separately, confirm the cable is included or purchase one — they are available from the same retailers listed above.

Operator Workstation — Bill of Materials (Per Station)

ITEM	MODEL / SPEC	WHERE TO BUY
— Standard Workstation (Pi 500) —		
Raspberry Pi 500 keyboard computer	Pi 500 standalone — Pi 5 chip, 8 GB RAM, 32 GB A2 microSD, keyboard	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor	15.6" Full HD IPS, built-in speakers, kickstand, VESA mount	raspberrypi.com GigaParts · Amazon
micro-HDMI to HDMI cable	1m micro-HDMI to standard HDMI (Pi 500 → Monitor) Included in Pi 500 Kit ($120) — verify before ordering separately	raspberrypi.com Amazon
USB mouse	Any USB or Bluetooth mouse — included in Pi 500 Kit	Amazon
USB-C power supply for Pi 500	Official Raspberry Pi 27W USB-C PD power supply (or equivalent)	raspberrypi.com Amazon
USB-C power supply for monitor (optional, for full brightness)	5V/1.5A USB-C supply — skip if powering monitor from Pi USB port (Pi USB port gives 60% brightness which may be dim outdoors)	Amazon
— Premium Workstation (Pi 500+) —		
Raspberry Pi 500+ keyboard computer	Pi 500+ — Pi 5 chip, 16 GB RAM, 256 GB NVMe SSD, mechanical keyboard (Gateron Blue), RGB	raspberrypi.com Adafruit · Amazon
Raspberry Pi Monitor + cables + mouse	Same as Standard Workstation above	Same as above

■ **501(c)(3) Grant Funding Note:** MCESV/MCEMA is a 501(c)(3) nonprofit. Operator workstation equipment may be eligible for funding through: FEMA BRIC (Building Resilient Infrastructure & Communities), ARPA-E Emergency Communications grants, ARRL Foundation grants, local Community Foundation equipment grants, or AUXCOMM/ARES regional funding. Document each workstation as "emergency communications operator terminal for the McHenry County Emergency Services Volunteers (K9ESV), a 501(c)(3) nonprofit supporting McHenry County Emergency Management Agency (MCEMA) RACES/ARES/Starcom communications."

Recommended Deployment by Position

POSITION	RECOMMENDED DEVICE	REASON
Net Control Operator (fixed EOC desk)	**Pi 500 + Pi Monitor**	Dedicated keyboard, good display, Chromium runs every FieldComms page smoothly
ICS Section Chief (Planning, Ops, Logistics)	Pi 500 + Pi Monitor or Windows/Mac laptop	Needs reliable keyboard for ICS form entry; Pi 500 is sufficient for all ICS platform pages
Served Agency Staff (EOC or shelter)	Any tablet, Chromebook, or Pi 500	Observer Mode and ICS forms work on any browser — no special hardware needed
Winlink / JS8Call Operator	Windows laptop (required)	Winlink Express and JS8Call are Windows applications — Pi 500 cannot run them natively

Mobile / Field Operator	Phone or tablet	FieldComms works on any smartphone browser on EMCOMM-NET — no dedicated hardware needed

■ The Pi 500 cannot run Windows applications natively. Winlink Express and JS8Call must run on the Windows laptop connected to the IC-7300. The Pi 500 is ideal for all browser-based FieldComms tools — net control logging, ICS forms, tactical maps, resource tracking, and the full ICS platform.